

Hand-in 1: Mar 10, 2026

Product Development and Engineering Design 151-3202-00L

Group 7

Authors:

Luis Casado

Alexandre Sanmartin

Antoine de Vinck

Eric Pizem

Jonas Harbarth

Exercise 1

- **Develop two interview questions: one question for a like/dislike interview and one question for an articulated interview.**

(Luis) Like/dislike method: Which are the limiting factors that prevent increasing planting capacity? For example: fatigue, time constraints, weather conditions or other factors?

(Luis) Articulated method: Can you explain how you clean the machine and how frequently you do it?

(Antoine) Like/dislike method: As you are planting seeds entirely by hand, what specific aspects of doing it manually do you enjoy and hate the most?

(Antoine) Articulated method: Walk us through a typical day of manual planting, all the way through your physical process in the field, to when you clean up and store.

(Jonas) Like/dislike method : what do you think is efficient/inefficient when planting ?

(Jonas) Articulated method : what part of the process can be most optimised ?

(Eric) Like/dislike method: what do you like/dislike on the tool you are currently using ?

(Eric) Articulated method: what are the main issues you would like to fix ?

(Alex) Like/dislike method: What do you like about the current devices and would like to conserve in the future generation?

(Alex) Articulated method: What type of people or user it's going to use and operate this machine?

- Carry out initial background research on project topic. What solutions currently exist? What are their strengths and weaknesses?



Figure 1: Different methods for the Specific Farming Use Case

Existing Solutions	Type of Use	Pros	Cons	Example
Manual Jab Planter	Manual Work, Distribute seeds and fertilizer Allows standing up work, Manual sticking in the ground and opening the hole.	Uses no Fuel, No extra infrastructure, Easy to use and store, Small soil disturbance (ridges conserved), Easy to repair and maintain.	Slow rate of planting, Manual and exhausting work with upper body, Heavy to carry, Limited load capacity,	Krupp 13A Manual Jab Planter
Animal-Drawn Direct Seeder	Pulled by oxen or donkeys, Cutting disk slicing the	Very effective, Not tiring and not demanding of human effort,	Need of animals and the extra costs these carry, Might mess up	

	terrain,	Simple and easy machinery, Easy to repair and maintain.	the ridges of the soil,	
Heavy Machinery (High Power Tractors and their attachments)	Machine-type of use, Single or multi user operated, Requires fuel, infrastructure, Needs special attachments to fulfill the functions	Very fast and efficient for big surfaces, High power and autonomy, Does not require human physical effort, Optimal precision achiever, Regulable modes of operation.	Needs fuel. Specialist required in case of damage of the machine. Too heavy to carry. Superficial fertilization. Complex machinery. Difficult to store and maintain correctly	Attachment: Highboy application or fertilizers like the ones from Zeppelin Maquinaria

Table 1: Comparison and Pros/Cons of the existing solutions for the current problem

Exercise 2

Target User 1: Farmer in Malawi

- Who is the user?
A smallholder farmer in Malawi cultivating about 4 Ha of land using manual tools.
- What do they need?
A cheap, fully manual precision seeder.
- Why do they need it?
 1. To improve planting precision
 - a. Correct depth (avoid planting deeper than rooting zone)
 - b. Proper covering (prevent seeds drying out).
 - c. Avoid fertilizer burning seeds.
 - d. Reduce replanting (which happens 2–3 days later if germination fails)
 2. Reduce physical effort
 - a. The current method causes back pain by leaning forward.
 - b. Arm strain from caring for the seeds.
 - c. Sticky clay soil requires excessive force to remove the hoe
- How is the task or activity currently done?
 1. Holes are dug manually on top of the ridge using a hoe. 10-30 cm depth.
 2. Holes are separated around 25cm
 3. Seeds are dropped by hand into each hole
 4. Seeds are covered using the foot, carefully avoiding ridge compression
- Where does it take place?
In ridge-based fields in rural Malawi. Fields are at least 1km from the village. Farmers walk or bike to the fields.
- When does it take place?
Ridges are prepared between October-November.
Entire field must be planted within one day
Replanting, if needed, happens 2-3 days after initial planting

Target User 2: Garden Owner

- Who is the user?
Garden owner who wants to plant rows of crops or flowers in their personal property for fun/hobby. Wants to make it fast and efficient, and not physically tiring. Also applies for people who want to grow their own produce to create byproducts (liquor, bread...)
- What do they need?
Simple cheap device that does not require the user to bend over continuously and reduce the time needed for new crops and new plantations.

Easy to store, repair and maintain for the sake of simplicity, since it is a for-fun activity.

- Why do they need it?
Make their hobby less tiring and more accessible for older people. Reduce physical strain and the time consumed in the process.
- How is the task or activity currently done?
Manually with the labor of more than one person (family and friends), and creating irregular patterns and distances.
- Where does it take place?
Any mid size property across the world where the owner cares about the garden or wants to produce their own goods.
- When does it take place?
Depends on the desired crops and location on the planet.

Exercise 3

3.1 Product Opportunity Gap

A Product Opportunity Gap is the space between what current products offer and what users actually need. In the context of small-scale farming in Malawi, this gap becomes visible when comparing existing planting methods with the practical constraints faced by local farmers.

Smallholder farmers in Malawi predominantly rely on manual seed planting methods. Seeds are typically sown by hand, a process that is inexpensive but labor-intensive, time-consuming, and often imprecise. Uneven seed spacing and inconsistent planting depth can reduce germination efficiency and lead to unnecessary seed waste, directly affecting crop yield and productivity.

At the same time, various low-cost manual precision planters and mechanized seed drills are available on the global market. These tools improve planting accuracy and reduce physical strain. However, many of these solutions are either financially inaccessible for farmers cultivating approximately one hectare, or they depend on manufactured components and spare parts that are not readily available in rural Malawian contexts. In many cases, replacement parts must be imported, leading to additional costs, long delivery times, and extended equipment downtime.

For small-scale farmers operating under tight financial constraints, this logistical dependency represents a significant barrier. Even when the initial purchase price is manageable, limited access to maintenance infrastructure and spare parts reduces the long-term viability of existing solutions.

Therefore, a structural gap persists between affordable but imprecise manual sowing and efficient yet economically and logistically inaccessible precision planting systems.

This gap highlights the opportunity to develop a manually operated precision seed-planting device that improves planting accuracy and reduces physical strain while remaining affordable, robust, and fully repairable using locally available materials and simple mechanical principles.

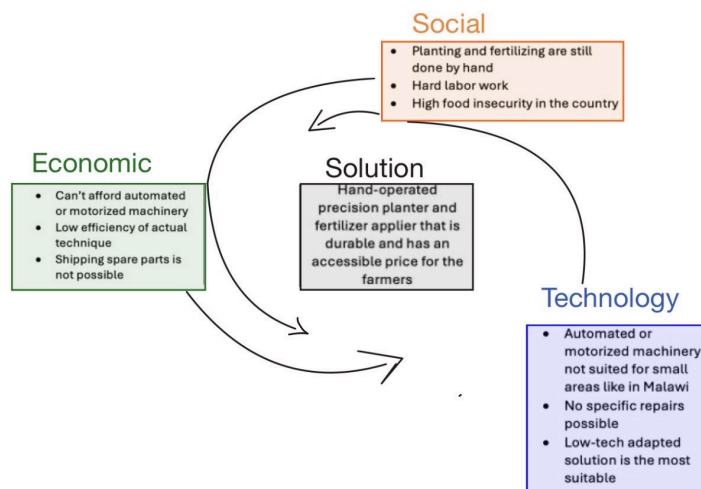


Figure 2: Summary graph of the statements described above with the three categories: Social, Technology, and Economic

3.2. Target user story

Frank, a smallholder farmer in his early 40s in a rural village in central Malawi, wakes up early as the soil conditions are perfect for planting. His one-hectare field, prepared with well-shaped ridges, is ready after weeks of work. Like many farmers in the region, he uses manual tools and traditional farming methods. His fields are located about one kilometer away from his village, so early in the morning he walks or cycles to the fields carrying his tools and seeds.

It is the beginning of the planting season around early December. The ridges have been done between October and November.

Once he arrives at the crops, he fills his manual precision planter with seeds and fertilizer and positions it at the start of the first ridge. As he pushes it forward, the tool creates a small hole, drops 1 to 3 seeds at a controlled depth, between 10 to 30cm, with 25cm separation between the holes and places fertilizer 5 to 10 cm beside them. The mechanism then covers the soil automatically, so Frank does not need to bend down.

He moves steadily along each ridge, repeating the same process. The spacing remains even, and the fertilizer is applied precisely next to every planting point.

After several hours, the entire field is seeded and fertilized uniformly. With rain expected soon, Frank is confident that the seeds will germinate evenly and grow under good conditions.

When he arrives back home he folds the precision seeder and stores it in a corner of his house. He's very happy because the seeder is almost not dirty and he doesn't need water to clean it.

3.3 Mission Statement

Product Opportunity Gap	Hand-operated precision planter and fertilizer applier that is durable and has an accessible price for the farmers
whBenefit Proposition	• Ensures correct seed planting depth and consistent spacing. • Place fertilizer safely beside seeds. • Reduces physical strain from bending and repetitive digging. • Improves germination rates and reduces replanting.
Key Business Goals	• Design a low-cost manual precision seeder, easy to use and durable. • Develop a functional prototype within the project timeline.
Target Market	• Smallholder farmers in Malawi relying primarily on manual agricultural tools.
Secondary Market	• Smallholders farmers in other countries with similar soil conditions. • NGOs supporting rural farming. • Garden owners.
Assumptions and Constraints	• Fully manual operation (no fuel or electricity). • Must function in ridge-based planting systems. • Must operate effectively in heavy or sticky clay soils. • Must remain affordable and easy to repair locally.
Stakeholders	• Farmers. • Manufacturers and distributors. • Designers and engineers involved in the project.

Table 2: Mission Statement- Key Statements of the Opportunity GaP

3.4. Pros and cons of using previous steps

Pros

- Provide a structured understanding of farmers' practical constraints (labor intensity, time pressure, and precision needs).
- Help define clear functional objectives for the machine design.
- Support the translation of contextual observations into measurable product requirements.

Cons

- Absence of field validation or participatory design process.
- Limited direct user feedback, as such devices are not widely used in Malawi.
- The target user story is based on assumptions and may not fully reflect real farming practices.
- Risk of subjective bias in interpreting farmers' needs.

3.5. User needs

To ease the following discussion, the acronym **PSF** (Precision Seeder Fertilizer) is used.

3.5.1 Categorized User Needs

The following user needs were derived from interview statements and contextual analysis. They are expressed in functional terms rather than technical solutions.

1. Fertilizer Application, Planting & Seed Spacing

1. The PSF increases the overall efficiency of the seeding and fertilizing process.
2. The PSF ensures fertilizer holes are properly covered to reduce nutrient losses (e.g., ammonia volatilization).
3. The PSF guarantees constant seed spacing and accurate dosage without missed deposits.
4. The PSF maintains consistent planting depth across different soil conditions.

2. Machine Design, Handling & Usability

5. The PSF allows sufficient seed and fertilizer capacity for continuous operation.
6. The PSF allows good visibility and control of the seeding and fertilizing process
7. The PSF requires limited physical effort to move forward on typical rural soils.
8. The PSF features a compact design suitable for household storage.
9. The PSF can be operated by a single user.
10. The PSF operates without requiring electricity or complex infrastructure
11. The PSF can be learned quickly by users with minimal training
12. The PSF enables easy filling and emptying of seeds and fertilizer.
13. The PSF allows intuitive and simple parameter adjustments.

3. Field Adaptability & Efficiency

14. The PSF allows adjustment of seed and fertilizer quantities to match different planting strategies and crops.
15. The PSF adapts geometrically to ridge-based planting systems.

4. Transport

16. The PSF enables easy transportation to and from the field.

5. Durability & Repairability

17.The PSF requires minimal cleaning time after use.

18.The PSF can be repaired and maintained using locally available tools and spare parts.

19.The PSF operates reliably without frequent clogging or malfunction.

6. Cost

20.The PSF represents an affordable investment for smallholder farmers.

3.5.2 Importance Ranking – Numerical Rating Method

The importance of each user is evaluated using a **numerical rating scale from 1 to 5**, where:

- **5 = Critical** (directly affects productivity, adoption, or economic viability)
- **4 = Very Important** (strong influence on usability or reliability)
- **3 = Important** (improves performance but not decisive)
- **2 = Desirable** (adds convenience but limited strategic impact)
- **1 = Low Priority** (minor influence on overall product success)

The ranking is based on interview emphasis, expected agronomic impact, usability considerations, and local economic constraints.

Level 5 – Critical

- Increased efficiency of the seeding/fertilizing process (1)
- Constant spacing and accurate dosage (3)
- Consistent planting depth (4)
- Limited physical effort required during operation (7)
- Repairability with locally available tools and spare parts (18)
- Affordable solution (20)
- Reliable operation without frequent clogging or malfunction(19)

These needs are essential for ensuring productivity, economic feasibility, and long-term adoption.

Level 4 – Very Important

- Fully mechanical operation (10)

- Proper fertilizer hole coverage (2)
- Push mechanism for improved visibility and control (6)
- Adjustable seed and fertilizer quantities (11)

These significantly influence reliability and adaptability in field conditions.

Level 3 – Important

- Adequate capacity for continuous work (5)
- Adaptability to ridge geometry (14)
- Easy filling and emptying (12)
- Single-operator use (9)
- Adaptability to ridge geometry (15)

These improve usability and efficiency but are not determining factors.

Level 2 – Desirable

- Minimal cleaning time (17)
- Easy transportation (16)
- Intuitive parameter adjustment (13)

These enhance user convenience but do not directly affect core performance.

Level 1 – Low Priority

- Compact design for storage (8)

This requirement has limited impact on functionality and adoption compared to agronomic and economic factors.

3.5.3 Reflection on the Approach

Pros: It is simple and easy to apply. It allows a clear overview of the user needs and helps organize priorities in a structured way. It also supports design decisions in an understandable manner.

Cons: The ranking remains partly subjective and is not fully data-driven. The limited number of priority levels can group several needs within the same category, even if their real importance differs. Additionally, the method depends strongly on the quality and representativeness of the interviews.

3.6 Activity Diagram

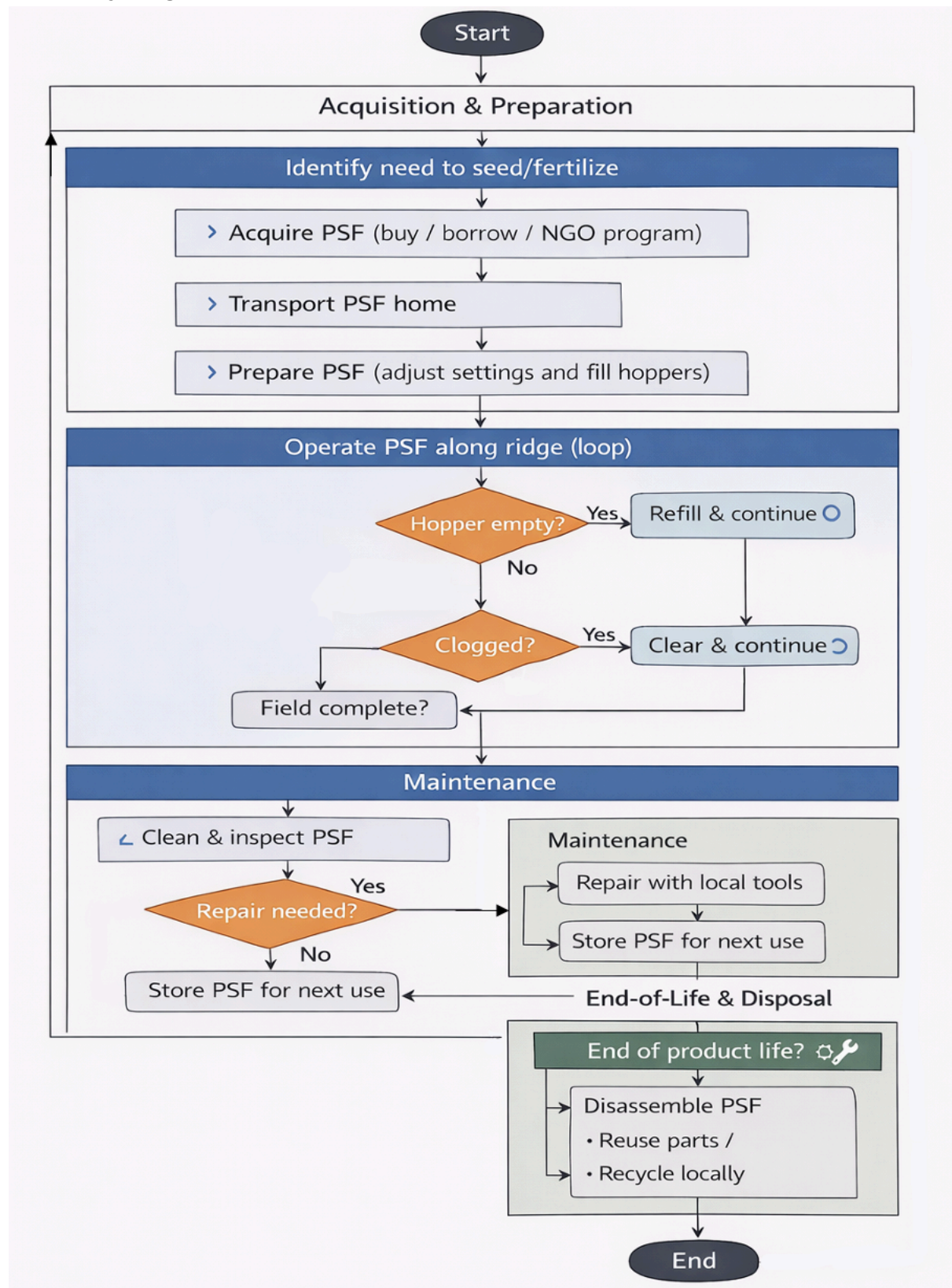


Figure 3: Activity Diagram of the Product

3.7Needs-Metrics Matrix

3.7.1. List of metrics

Metric No.	Need No.	Cat.	Metric	Imp.	Units
1	1	1	Daily field coverage	5	ha/h
2	2	1	The seed is covered	5	nr of covered seeds
3	3	1	Distance between planting stations	5	cm
4	3	1	Number of seeds placed per station	5	nr of seeds
5	3	1	Distance between seed and fertilizer	5	cm
6	4	1	Hole depth created	5	cm
7	7	2	Number of bending motions required per hour	5	count
8	10	2	External fuel/energy required	5	binary pass/fail
9	19	5	Cycles to failure in harsh soil conditions	5	cycles
10	20	6	Target unit retail cost	5	US \$
11	3	3	Planting accuracy (percentage of seeds placed at exact target depth and spacing)	5	%
12	3	1	Mass of fertilizer dispensed per station	4	g
13	7	2	Total mass of the empty planting system	4	kg
14	9	2	Number of operators required	4	nr of people
15	14	3	Ground pressure applied to soil ridges	4	kPa
16	18	5	Number of moving parts	4	count
17	18	5	Mass of soil adhered to the system after 1 hour in clay soils	4	g
18	20	6	Cost of replacement parts	4	US \$
19	5	2	Seed hopper maximum weight capacity	3	kg
20	15	4	Dimensions for transport (Length × Width × Height)	3	cm
21	16	5	time required for cleaning	3	Min/day of use
22	18	5	Percentage of standard/locally available components	3	%
23	1	3	Overall planting precision while maintaining target speed	3	%
24	8	2	Storage volume required	2	cm ³
25	13	1	Changing distance between planting stations	2	operation range in cm

Table 3: Metrics

1	Fertilizer Application,Planting & Seed Spacing
2	Machine Design, Handling & Usability
3	Field Adaptability & Efficiency
4	Transport
5	Durability & repairability
6	Cost

Table 4: Categorization Criteria

No	Cat	User Need / Engineering Requirement	Daily field coverage	Distance between planting stations	Fertilizer quantity delivered per seed	Separation distance between fertilizer and seed	Number of seeds deposited per planting station	Hole depth created	Target unit retail cost	Required pushing force on the planter	Number of operators required	Percentage of standard/locally available components	Cleaning duration	Maximum capacity for seeds and fertilizer	Ability to adjust seed and fertilizer dosage	Total weight of the planting system	Maximum overall length
1	1	The PSF increases the overall efficiency of the seeding and fertilizing process.	X														
2		The PSF ensures fertilizer holes are properly covered to reduce nutrient losses		X													
3		The PSF guarantees constant seed spacing and accurate dosage without missed deposits			X	X	X									X	
4		The PSF maintains consistent planting depth across different soil conditions						X									
5	2	The PSF allows sufficient seed and fertilizer capacity for continuous operation												X			
6		The PSF allows good visibility and control of the seeding and fertilizing process		X													
7		The PSF requires limited physical effort to move forward on typical rural soils								X						X	
8		The PSF features a compact design suitable for household storage														X	
9		The PSF can be operated by a single user									X						
10		The PSF operates without requiring electricity or complex infrastructure															
11		The PSF can be learned quickly by users with minimal training														X	
12		The PSF enables easy filling and emptying of seeds and fertilizer														X	
13	3	The PSF allows intuitive and simple parameter adjustments														X	
14		The PSF allows adjustment of seed and fertilizer quantities to match different planting strategies and crops													X		
15	4	The PSF adapts geometrically to ridge-based planting systems.															X
16		The PSF enables easy transportation to and from the field															X
17	5	The PSF requires minimal cleaning time after use											X				
18		The PSF can be repaired and maintained using locally available tools and spare parts										X					
19		The PSF operates reliably without frequent clogging or malfunction											X				
20	6	The PSF represents an affordable investment for smallholder farmers							X								

Table 5: Needs-Metrics Matrix

3.8. Target Product Specifications

Metric number	Corresponding need number	Category	Metric	Importance	Units	Marginal Value	Ideal Value	Responsible Team member	Update date
1	1	Field Adaptability & Efficiency	Planting speed	5	ha/h	≥ 0.20	≥ 0.40	Luis	09/03/2026
2	2	Fertilizer Application, Planting & Seed Spacing	Seed coverage rate	5	%	≥ 85	≥ 95	Luis	09/03/2026
3	3	Fertilizer Application, Planting & Seed Spacing	Distance between planting stations	5	cm	25 ± 5	25 ± 2	Luis	09/03/2026
4	3	Fertilizer Application, Planting & Seed Spacing	Number of seeds placed per station	5	count	$1-3 \pm 1$	Selectable {1;2;3} ± 0	Eric	09/03/2026
5	3	Fertilizer Application, Planting & Seed Spacing	Distance between seed and fertilizer	5	cm	5–10	8–10	A.V	09/03/2026
6	4	Fertilizer Application, Planting & Seed Spacing	Hole depth created	5	cm	10–30 adjustable	10–30 adjustable (± 1 cm control)	A.V	09/03/2026
7	7	Machine Design, Handling & Usability	Number of bending motions required per hour	5	count	≤ 60	≤ 10	Alex	09/03/2026
8	10	Machine Design, Handling & Usability	External energy required	5	binary	None	None	Jonas	09/03/2026
9	19	Durability & Repairability	Cycles to failure in harsh soil	5	cycles	$\geq 30,000$	$\geq 100,000$	Eric	09/03/2026
10	20	Cost	Target unit retail cost	5	USD	≤ 8	≤ 5	Jonas	09/03/2026
11	3	Field Adaptability & Efficiency	Planting accuracy (depth & spacing)	5	%	≥ 80	≥ 95	Jonas	09/03/2026
12	3	Fertilizer Application, Planting & Seed Spacing	Mass of fertilizer dispensed per station	4	g	$5-8 \pm 15\%$	$5-8 \pm 5\%$	Alex	09/03/2026
13	7	Machine Design, Handling & Usability	Total mass of empty system	4	kg	≤ 10	≤ 6	A.V	09/03/2026
14	9	Machine Design, Handling & Usability	Number of operators required	4	persons	≤ 2	1	Jonas	09/03/2026

15	14	Field Adaptability & Efficiency	Ground pressure on ridges	4	kPa	≤30	≤15	Jonas	09/03/2026
16	18	Durability & Repairability	Number of moving parts	4	count	≤25	≤15	Eric	09/03/2026
17	18	Durability & Repairability	Soil adhesion after 1h in clay	4	g	≤150	≤50	Luis	09/03/2026
18	20	Cost	Replacement parts cost per year	4	USD	≤3	≤1	Alex	09/03/2026
19	5	Fertilizer Application, Planting & Seed Spacing	Seed hopper max capacity	3	kg	≥1	≥2	A.V	09/03/2026
20	16	Transport	Transport dimensions (L×W×H)	3	cm	≤120×60×60	≤100×50×50	Eric	09/03/2026
21	17	Field Adaptability & Efficiency	TIME required for cleaning	3	Min /day of use	≤30	≤20	Eric	09/03/2026
22	18	Durability & Repairability	Locally available components	3	%	≥70	≥90	Alex	09/03/2026
23	1	Field Adaptability & Efficiency	Overall planting precision at target speed	3	%	≥80	≥95	Luis	09/03/2026
24	8	Machine Design, Handling & Usability	Storage volume required	2	dm³	≤120	≤60	Alex	09/03/2026
25	13	Fertilizer Application, Planting & Seed Spacing	Adjustable planting spacing range	2	cm	50–90	25–100	A.v	09/03/2026

Table 6: Product Specifications

3.8.2. Reflect on the pros and cons

This table serves as general guidelines in the design and dimensioning of the product.

Pros : This target product specification gives an overview of the product design and capabilities. It is a clear way to categorize the specification and keep an order of magnitude behind any design choice that has to be made.

Cons : This method is clearly subjective and lacks data driven choice. As the ideal and marginal values are either taken from competitors' design, either from or subjective point of view, all of the values will have to be adjusted after first trials.

Exercise 4

Exercise 1: Antoine and Luis, Jonas

Exercise 2: Luis and Alexandre, Eric

Exercise 3: All